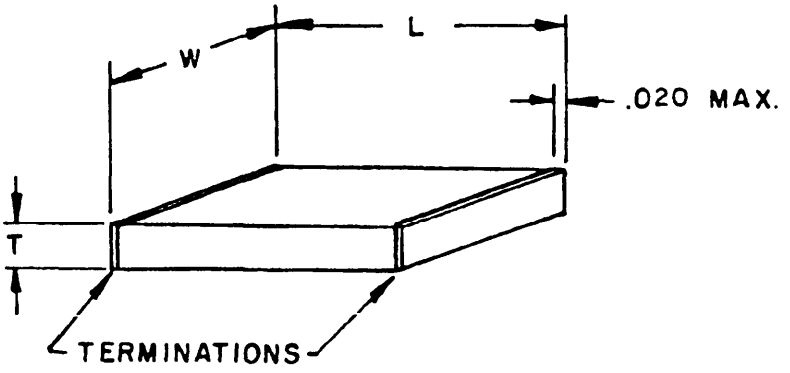
MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY STYLES CDR26, CDR27, CDR28, CDR29, AND CDR30

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-55681.



Style	Dimensions			
	L	W	T	
			Min.	Max.
CDR26	.150 ±.015	.150 ±.015	.060	.120
CDR27	.190 ±.019	.190 ±.019	.060	.120
CDR28	.330 ±.033	.330 ±.033	.060	.120
CDR29	.400 ±.040	.400 ±.040	.060	.120
CDR30	.540 ±.054	.400 ±.040	.060	.120

INCHES	MM
.015	.38
.019	.48
.020	.51
.033	.84
.040	1.02
.054	1.37
.060	1.52
.120	3.05
.150	3.81
.190	4.83
.330	8.38
.400	10.16
.540	13.72

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

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TABLE I. Styles CDR26, CDR27, CDR28, CDR29, and CDR30 capacitor characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR26BP151J---	150	F, G, J	BP	4000
CDR26BP181H---	180			3000
CDR26BP221H---	220			3000
CDR26BP271H---	270			3000
CDR26BP331H---	330			2000
CDR26BP391H---	390			↓
CDR26BP471H---	470			
CDR26BP561H---	560			
CDR26BP681F---	680			1000
CDR26BP821F---	820			
CDR26BP102F---	1000			
CDR26BP122F---	1200			
CDR26BP152F---	1500			
CDR26BP182F---	1800			
CDR26BP222F---	2200			↓
CDR27BP271J---	270			4000
CDR27BP331J---	330			4000
CDR27BP391H---	390			3000
CDR27BP471H---	470			3000
CDR27BP561H---	560			3000
CDR27BP681G---	680			2000
CDR27BP821G---	820			↓
CDR27BP102G---	1000			
CDR27BP122G---	1200			
CDR27BP272F---	2700			1000
CDR27BP332F---	3300			↓
CDR27BP392F---	3900			
CDR27BP472F---	4700			
CDR28BP391J---	390			4000
CDR28BP471J---	470			↓
CDR28BP561J---	560			
CDR28BP681J---	680			
CDR28BP821H---	820			3000
CDR28BP102H---	1000			3000
CDR28BP122H---	1200			3000
CDR28BP152G---	1500			2000
CDR28BP182G---	1800			↓
CDR28BP222G---	2200			
CDR28BP272G---	2700			
CDR28BP562F---	5600			1000
CDR28BP682F---	6800			↓
CDR28BP822F---	8200			
CDR28BP103F---	10000			
CDR28BP123F---	12000			
CDR29BP821J---	820			4000
CDR29BP102J---	1000			4000
CDR29BP122J---	1200			4000
CDR29BP152H---	1500			3000
CDR29BP182H---	1800			3000
CDR29BP222H---	2200			3000
CDR29BP332G---	3300			2000
CDR29BP392G---	3900			↓
CDR29BP472G---	4700			
CDR29BP562G---	5600			↓
CDR29BP153F---	15000			1000
CDR29BP223F---	22000			1000
CDR30BP152J---	1500	↓	↓	4000
CDR30BP182J---	1800			4000

See footnote at end of table.

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TABLE I. Styles CDR26, CDR27, CDR28, CDR29, and CDR30 capacitor characteristics - Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR30BP222J---	2200	F, G, J	BP	4000
CDR30BP272H---	2700	↓	↓	3000
CDR30BP332H---	3300			3000
CDR30BP682G---	6800			2000
CDR30BP822G---	8200			2000
CDR30BP273F---	27000			1000
CDR30BP333F---	33000	↓	↓	1000
CDR26BX151J---	150	K, M	BX	4000
CDR26BX181J---	180	↓	↓	↓
CDR26BX221J---	220			
CDR26BX271J---	270			
CDR26BX331J---	330			
CDR26BX391J---	390			
CDR26BX471J---	470			↓
CDR26BX561H---	560			3000
CDR26BX681H---	680			3000
CDR26BX821H---	820			3000
CDR26BX102G---	1000			2000
CDR26BX122G---	1200			↓
CDR26BX152G---	1500			
CDR26BX182G---	1800			
CDR26BX222G---	2200			↓
CDR26BX272F---	2700			1000
CDR26BX332F---	3300			↓
CDR26BX392F---	3900			
CDR26BX472F---	4700			
CDR26BX562F---	5600			
CDR26BX682F---	6800			
CDR26BX822F---	8200			
CDR26BX103F---	10000			↓
CDR27BX561J---	560			4000
CDR27BX681J---	680			↓
CDR27BX821J---	820			
CDR27BX102J---	1000			↓
CDR27BX122J---	1200			3000
CDR27BX152H---	1500			3000
CDR27BX182H---	1800			2000
CDR27BX272G---	2700			↓
CDR27BX332G---	3300			
CDR27BX392G---	3900			
CDR27BX472G---	4700			
CDR27BX562G---	5600			↓
CDR27BX123F---	12000			1000
CDR27BX153F---	15000			1000
CDR27BX223F---	22000			1000
CDR28BX152J---	1500			4000
CDR28BX182J---	1800			4000
CDR28BX222J---	2200			4000
CDR28BX272H---	2700			3000
CDR28BX332H---	3300			3000
CDR28BX392H---	3900			3000
CDR28BX682G---	6800			2000
CDR28BX822G---	8200			↓
CDR28BX103G---	10000			
CDR28BX123G---	12000			
CDR28BX273F---	27000			1000
CDR28BX333F---	33000			↓
CDR28BX393F---	39000			
CDR28BX473F---	47000	↓	↓	↓

See footnote at end of table.

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TABLE I. Styles CDR26, CDR27, CDR28, CDR29, and CDR30 capacitor characteristics - Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR29BX272J---	2700	K,M	BX	4000
CDR29BX332J---	3300			4000
CDR29BX392J---	3900			4000
CDR29BX472M---	4700			3000
CDR29BX562M---	5600			
CDR29BX682M---	6800			
CDR29BX822M---	8200			
CDR29BX153C---	15000			2000
CDR29BX223G---	22000			2000
CDR29BX563F---	56000			1000
CDR29BX823F---	82000			1000
CDR30BX103M---	10000			3000
CDR30BX123M---	12000			3000
CDR30BX273G---	27000			2000
CDR30BX333G---	33000			2000
CDR30BX104F---	100000			1000
CDR30BX124F---	120000			1000
CDR30BX154F---	150000			1000

1/ Complete type designation will include additional symbols to indicate tolerance, termination finish, and failure rate level.

REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Case type: Multi-layer, unencapsulated, monolithic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Marking: In accordance with MIL-C-55681.

Voltage conditioning: In accordance with MIL-C-55681 except that 120 percent of the rated voltage shall be applied to the unit at the maximum rated temperature $^{+4}_{-0}$°C for 100 ±4 hours.

Insulation resistance: In accordance with MIL-C-55681 except that for the test conditions 1000 volts shall be applied through a resistor sufficient to limit the charging current to a maximum of 50 milliamperes (mA).

Dielectric withstanding voltage: In accordance with MIL-C-55681 except that the magnitude and nature of test voltage shall be 120 percent of rated voltage.

Voltage-temperature limits: In accordance with MIL-C-55681 except that steps E, F, and G shall be deleted from table X voltage-temperature limit cycle.

Life: In accordance with MIL-C-55681 except that for the operating conditions, capacitors shall be subjected to 120 percent of rated voltage. The surge current shall not exceed 50 mA. When necessary, a suitable current-limiting resistor shall be inserted into the current.

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Quality factor: Not applicable.

Resonance: Not applicable.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - AS, OS
Air Force - 17, 85
DLA - ES

User activities:
Navy - MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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